

Mars Reconnaissance Orbiter: Extended Dual-Purpose Mission

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Abstract—The Mars Reconnaissance Orbiter (MRO) was launched from Cape Canaveral Air Force Station, Florida, aboard an Atlas V-401 launch vehicle on August 12, 2005 and entered Mars orbit on March 10, 2006. After five months of aerobraking, a series of propulsive maneuvers were used to establish the desired low-altitude science orbit. The spacecraft has been on station in its 255 x 320 km, 3 p.m., primary science orbit since September 2006 performing its scientific and UHF-relay functions. This paper provides a brief status of the prime mission and describes the plans and preparations in work for its recently approved two-year mission extension^{1 2}

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1. INTRODUCTION

The Mission

The Mars Reconnaissance Orbiter (MRO) spacecraft is currently in orbit at the planet Mars, providing unique scientific data unprecedented in quantity and quality. As described in a previous papers^[1,2], MRO is a flagship mission sent to Mars as part of the strategic plan promulgated by NASA's Mars Exploration Program. The Program is designed to a common theme – “Follow the Water”, and with four strategic elements:

- Search for evidence of past or present life;
- Understand the climate and volatile history of Mars;
- Understand the geology and geophysics of the

Martian surface and subsurface; and

- Assess the nature and inventory of resources on Mars in anticipation of human exploration.

MRO supports this program in two different ways. First, MRO directly addresses the above four strategic elements by using its instrument suite to:

- Characterize Mars' seasonal cycles and daily variations of water, dust & carbon dioxide;
- Understand the climate and volatile history of Mars;
- Characterize Mars' global atmospheric structure, transport and surface changes;
- Search sites for evidence of aqueous and/or hydrothermal activity;
- Characterize in detail the stratigraphy, geology & composition of Mars surface features;
- Characterize the Martian ice caps and the polar layered terrains;
- Profile the upper crust while probing for subsurface water and ground ice;
- Identify and characterize many sites for future landed missions.

Second, MRO provides programmatic support by:

- Identification, characterization, and certification of landing sites for landed assets (rovers and landers), and;
- UHF relay support to landed assets, both for critical events like Entry, Descent, and Landing (EDL) and for nominal science and engineering data relay.

The Vehicle

The MRO Project was designed from the first to raise the bar in orbiting capabilities for Mars. NASA selected a scientific payload of instruments that were to provide an order of magnitude increase in spatial and spectral resolution at Mars in many parts of the electromagnetic spectrum. To achieve this, the Jet Propulsion Laboratory (JPL) chose to partner with Lockheed Martin Space Systems Company (LMSSC) to design, build, test and fly a vehicle capable of operating in a lower altitude orbit around Mars, of combining global monitoring, regional survey, and

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coordinated targeting, and of returning an order of magnitude increase in data volume.

The spacecraft, seen in Figure 1, consists of a highly capable orbiter bus, a suite of 6 instruments, and 3 engineering payloads integrating flight heritage with new technology capabilities.

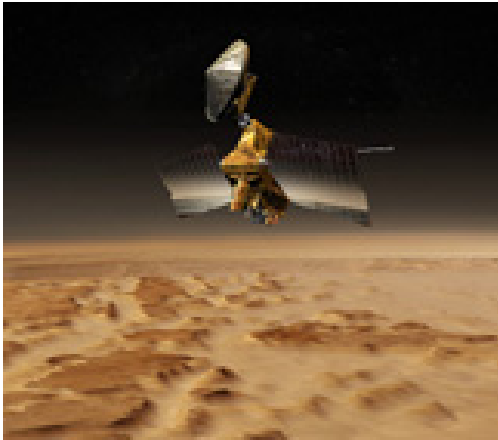


Figure 1. Artist rendering of the MRO spacecraft

The orbiter bus has a significant set of advances over previous missions to Mars, including:

- The Electrical Power Subsystem (EPS) – The solar energy conversion system capable of producing over 2.2 kilowatts at Mars, allowing full use of all instruments without power management constraints.
- The capability to transmit up to 6 Mbits per second downlink, enabled by a 3 meter High Gain Antenna and a 100 watt transmitter.
- A software algorithm (ITL – Integrated Target List) allowing for the flight team to command experiment observations in latitude and longitude of the observation, allowing the spacecraft to autonomously adjust the timing as required by updated spacecraft ephemerides to achieve precise pointing.
- Large reaction control wheels and coupled thrusters – Minimizes the spacecraft induced disturbances into navigation predicts.

Utilizing all this new capability, the instrument suite includes:

- HiRISE, the High Resolution Imaging Science Experiment. Provided by the University of Arizona (PI: Alfred McEwen), key features include a spatial resolution as high as 30 cm/pixel in a swath 6 km wide in a panchromatic visible band and in a 1.2 km wide swath in blue-green and near-IR bands for color. High signal to noise is achieved by the use of

time-delayed-integration (TDI).

- CRISM, the Compact Reconnaissance Imaging Spectrometer. Provided by the Johns Hopkins University Applied Physics Laboratory (PI: Scott Murchie), key features include visible and near-IR detectors combining for 512 spectral channels from 0.37 to 3.9 microns. This spectral information is achieved in images ~11 km across and with highest spatial resolution of ~18 m/pixel.
- SHARAD, the Shallow subsurface Radar. Provided by ASI, the Italian Space Agency and INFOCOM at the University of Rome (TL: Roberto Seu), key details include penetration depths of several hundred meters with a vertical resolution at the central frequency in free space of 15 m.
- MARCI, the Mars Color Imager. Provided by Malin Space Science Systems (PI: Michael Malin), key features include wide angle imaging from horizon to horizon orthogonal to the spacecraft track with a spatial resolution of ~ 1 km/pixel in 2 UV and 5 Visible bands.
- CTX, a context camera. Also provided by MSSS (TL: Michael Malin), key features include a spatial resolution of ~6 m/pixel in a 30-km swath.
- And MCS, the Mars Climate Sounder. Provided by the Jet Propulsion Laboratory (PI: Daniel McCleese), key features include 9 channels in visible and infrared, making measurements every 5 kilometers down through the atmosphere.

The advanced technology engineering payloads include:

- An Optical Navigation Camera (ONC), used to improve the ephemeris of the two small Martian moons, thereby pathfinding improved entry navigation for future landers.
- A Ka-band transmission system, used to flight verify key technological and operational features of this higher rate, lower power frequency band.
- An Electra UHF relay radio subsystem, used to provide relay services to landed assets, including operational support to the Phoenix lander in 2008, as well as added support to the Mars Exploration Rovers (MER) and, starting in 2012, to the Mars Science Laboratory (MSL). A software reconfigurable radio system developed by the Jet Propulsion Laboratory, key features include the ability to upgrade performance by reprogramming the device for new frequencies, new coding standards, and software corrections.

2. MISSION SUMMARY – LAUNCH TO THE PRIMARY SCIENCE ORBIT

MRO was launched in August 2005. After a seven-month interplanetary transit, the spacecraft fired its six main

engines and established a highly elliptical capture orbit at Mars. During the post Mars Orbit Insertion (MOI) early check-out period, four instruments acquired engineering-quality data. This was followed by five months of aerobraking operations. After aerobraking was terminated, a series of propulsive maneuvers were used to establish the desired low-altitude science orbit. The spacecraft has been on station in its primary science orbit since September 2006. A description of the transition from launch to the primary science orbit can be found in Reference [3].

Primary Science Orbit Description

The primary science orbit (PSO) has been designed to satisfy the science requirements of the mission. It has the following characteristics:

- Sun-synchronous ascending node at 3 p.m. LMST — daylight equatorial crossing (near polar inclination of 92.7 deg)
- Eccentricity and argument of periapsis that results in a low altitude orbit “frozen” over the south pole (periapsis altitude of 255 km, apoapsis altitude of 320 km, and an argument of periapsis of 270 deg); and
- Semi-major axis that produces a 17-day (short term) groundtrack repeat cycle (semi-major axis of 3775 km).

The 3 p.m. local mean solar time (LMST) provides a balance between the visual imaging instruments (HiRISE, CTX) which prefer surface lighting contrasts (shadowing) and CRISM which spectrally divides reflected sunlight into many channels in order to identify surface composition.

Because of the 3:00 p.m. LMST orbit orientation, the MRO spacecraft will experience a solar eclipse on each orbit. Additionally, on almost every orbit, the spacecraft will experience an Earth occultation, causing the orbiter to lose contact with the DSN. This loss has a noticeable effect on the downlink data volume capability of the mission.

The near-circular, near polar orbit provides repeatable altitudes at each latitude and comparable altitudes at all latitudes greatly simplifying mosaicking and comparison of data taken on different orbits. The “frozen” orbit condition results in a periapsis that remains stationary. With the periapsis location fixed over the south pole of Mars, a 65-70 km range between the periapsis and apoapsis altitudes above the surface results naturally due to the Martian gravity field. Variations in the spacecraft altitude above the Martian surface at specific latitudes, however, are limited to just a few kilometers.

Because of the different observation modes (global mapping and profiling, regional survey, and globally distributed targeting) of the science suite, the PSO has been designed to produce different groundtrack repeat cycles. There is a long-term repeat cycle that provides uniform, global

coverage of Mars with a fine grid of less than 5 km at the equator. This is the exact repeat of the groundtrack that will occur after 4602 orbits [359 days (349 sols)]. Unfortunately, such a precise spacing cannot be maintained due to Mars gravity variations and to atmospheric perturbations at the relatively low orbit altitude (~300 km). In addition to the long-term repeat cycle, there is a second, shorter-term repeat cycle that has been dubbed the targeting cycle. The targeting cycle, which repeats every 211 orbits [17 days (16.5 sols)], has a 31 km westward walk relative to any selected reference node. When combined with the spacecraft’s ability to roll left or right (cross-track) up to 30°, this provides the opportunity to view nearly any local area on Mars within 17 days. This greatly simplifies the observation planning, as there are multiple opportunities to see a target and repeated imaging for stereo can be accomplished in a seasonally short period of time (~month). Because of impacts to global mapping investigations, targeted observations with roll angles less than 10 degrees are preferred, but larger rolls can be accommodated and are typically required for stereo.

Due principally to atmospheric perturbations, regular orbit trim maneuvers are performed to control the groundtrack repeat patterns. Additional information about science planning and mission operations is available elsewhere^[4,5].

Mission Data Acquisition and Return

The MRO spacecraft carries an instrument suite that is capable of observing the surface at unprecedented levels of resolution. To provide the increased resolution, MRO has been designed to collect and return to Earth, 10 to 20 times more data than previous Mars missions. The MRO spacecraft has a maximum data storage capability of 160 Gigabits and a telecom system that is capable of transmitting at very high data rates, even at maximum Earth-Mars ranges. As of September 2010, MRO has returned over a 123 Tb of total science data through the Deep Space Network (DSN). Designed to communicate with the DSN via a direct X-band link, MRO has become one of the DSN’s highest volume users requiring two 8-hour tracks per day (34m antenna coverage).

Mission Phases

The Mars Reconnaissance Orbiter Mission is divided in time into distinct mission phases. Table 1 contains the phase definitions relevant to operational activities conducted from the primary science orbit as well as approximate phase durations.

3. HIGHLIGHTS FROM THE PRIME MISSION (PRIMARY AND EXTENDED SCIENCE PHASES)

From its start, the Mars Reconnaissance Orbiter (MRO) mission has had both scientific and programmatic

Table 1. MRO Mission Phase Descriptions
(from the Primary Science Orbit)

<i>Mission Phase</i>	<i>Duration</i>	<i>Description</i>
Primary Science Phase (PSP)	Two Years (Sep 2006 to Dec 2008)	Begins after achieving the primary science orbit and continues for at least one Mars year (687 Earth days), starting at an appropriate time following turn-on of the instruments. The orbiter will provide relay support to Mars missions launched in 2007.
Relay & Extended Science Phase (ESP)	Two Years (Jan 2009 to Sep 2010)	Begins after the Primary Science Phase ends. Performs science observations and provides UHF relay support for the arrival and surface operations of spacecraft in the international Mars exploration program; phase continues until the end of 2010.
First Extended Mission (EM)	Two Years (Oct 2010 to Sep 2012)	Begins after the Extended Science Phase ends. Performs science observations and provides UHF relay support for the arrival and surface operations of spacecraft in the international Mars exploration program; phase continues until the end of 2012.

objectives. Some, such as conducting a comparative navigation experiment with the Optical Navigation Camera, demonstrating high rate downlink capability with the Ka-band telecommunications subsystem, and acquiring data during the aerobraking phase to characterize upper-atmospheric structure were completed prior to the start of the Primary Science Phase (PSP) in November 2006. By December 2008 the Mars Reconnaissance Orbiter had completed all of its Primary Science Phase mission success objectives and Level 1 requirements. Following solar conjunction in December 2008, MRO began an approved Extended Science Phase (ESP). This is running concurrently with a previously defined Relay Phase whose objectives were affected by the slip of the Mars Science Laboratory (MSL) launch from 2009 to 2011. Both ESP and the Relay Phase extend to December 31, 2010, the current end of mission for MRO. (Note: NASA has recently approved a mission extension for MRO. The end of the prime mission has been changed to September 26, 2010; the Extended Mission (EM) will extend from October 2010 to September 2012.)

Science Overview

Eight scientific investigations were selected for the Mars Reconnaissance Orbiter mission. Six of these provided instruments for flight: CRISM, CTX, HiRISE, MARCI, MCS and SHARAD. Two investigations, Gravity and Upper Atmospheric Structure (Accelerometer), used the spacecraft itself as the measurement tool. Daily tracking of the spacecraft using the X-band telecommunications system during its downlinking of data provides a sensitive measure of the gravity field and its variations due to the nonspherical distribution of mass on Mars. This includes two time-dependent components, one due to the condensation and sublimation of 25-30% of the atmosphere's total mass in the polar regions as temperatures fall and rise in the seasonal polar night and day. The other prominent temporal variation results from the strong diurnal heating of the lower atmosphere by radiation and convection and absorption of solar UV radiation in the upper atmosphere and propagation

of wave energy from lower altitudes. During the aerobraking phase of the mission, the spacecraft dipped into the atmosphere on each aeropass. As it did so, onboard accelerometers sensed the density variations that the spacecraft encountered. A more detailed description of the instruments and investigations can be found in Zurek and Smrekar^[6] and the references therein.

For MRO, it is convenient to divide discoveries into three time periods: Ancient Mars, Recent (geologically speaking) Mars, and Present Mars. Highlights from Ancient and Recent Mars have been discussed elsewhere^[7]. In this paper a brief overview and some updated results from observations of Present Mars is presented.

Science Highlights - Observations of Present Mars

To date, MRO CTX has covered more than 60% of the planet at its resolution of ~6 m/pixel; MRO HiRISE has covered 1% at its much higher spatial resolution of ~0.3 m/pixel. CRISM has covered ~75% of the planet during periods of low opacity mainly in its multi-spectral survey mode of 72 spectral channels at ~200 m/pixel, with about 15% of its coverage in a 254-channel hyper-spectral mode. Areas targeted by CRISM in full spectral (544 channels) and highest spatial mode (~18 m/pixel) cover about ~1% of Mars. (Note that the surface area of Mars is approximately the same as the land area of the larger Earth.)

Mars remains a dynamic planet with ongoing change. Impacts still occur today. More than 100 new impact craters have been identified by CTX as it compares against earlier CTX data and MGS MOC data. Many of the craters in northern mid-latitudes have exposed ice just below the surface. Once exposed the ice sublimates away at a rate suggesting relatively pure, dust-free deposits.

As shown in Figure 2, once spotted, HiRISE can follow up with very high resolution imaging to view the structure of the impacts and the nature of the material excavated, including how quickly the dark albedo features blend back

into the dust covered landscapes. Crater counting provides a check on the distribution curves used to estimate the relative ages of various parts of the Mars surface.

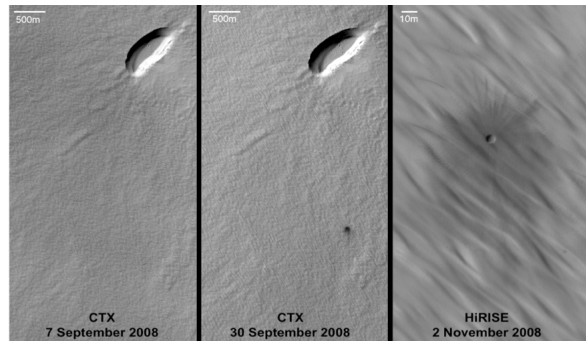


Figure 2. CTX and HiRISE imaging of new crater (Credit Malin Space Science Systems, University of Arizona, JPL, and NASA)

Other changes are more enigmatic: new gully features (7 to date) have emerged, but whether they are due to dry landslides or water-activated slurries continues to be debated. Figure 3 shows a recent material flow down a channel in the side of a crater and its dispersal out on the crater floor, moving around the local topography. It is unknown whether it is a dry avalanche or water aided flow. (Note: The bright material has the same spectral signature as dust on the crater wall—no signature of water-modified materials.)

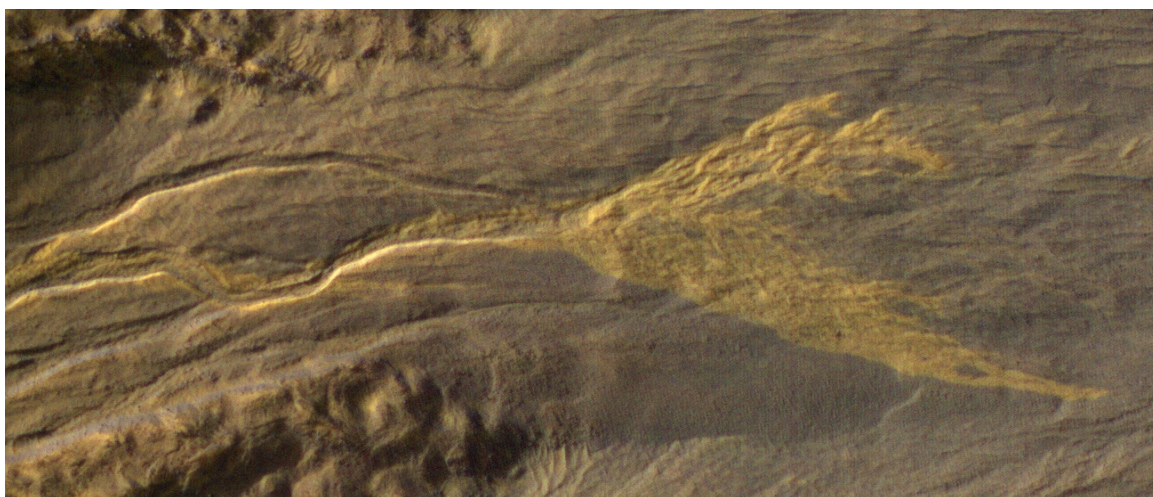


Figure 3. Recent material flow observed by HiRISE (Credit University of Arizona, JPL, and NASA)

Figure 4 shows a HiRISE image of a landslide at the edge of the permanent north polar cap triggered by springtime katabatic winds.



Figure 4. Dust avalanche imaged by HiRISE (Credit University of Arizona, JPL, and NASA)

Seasonal changes in the polar caps continue to occur and here the south polar area is particularly interesting with its thin layer of CO₂ ice cover lingering through summer and producing a variety of telltale signs of geyser-like activity. The warm soil sublimates the bottom of the CO₂ ice producing gas pockets which crack the overlying ice,

erupting dust-laden plumes which are blown downwind (Figure 5).



Figure 5. Geyser-like Phenomena imaged by HiRISE (Credit University of Arizona, JPL, and NASA)

One prominent change from year to year is the number and scale of dust storms on the planet. In 2007 a major dust event occurred on Mars, with dust hazes eventually obscuring most of the southern and part of the northern

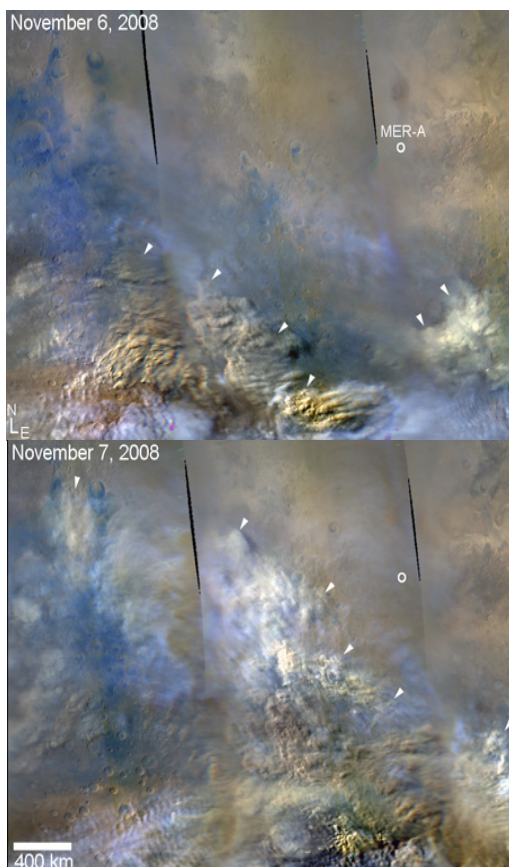


Figure 6. Dust Storms near the MER Rover Spirit captured by MARCI – Nov 2008 (Credit Malin Space Science Systems, JPL, and NASA)

hemisphere. MARCI observations combined with earlier MGS MOC observations now provide a decade long record (5 Mars years) of such events. MRO MARCI daily global maps of atmospheric dust activity also provided alerts to the Phoenix lander and to the two Mars Exploration Rovers, all solar-powered spacecraft. MARCI data were credited as being decisive in keeping the Spirit rover alive: warnings of an approaching local storm enabled implementation of a power-conservative mode before the storm arrived (Figure 6).

UHF Relay Highlights - 2007 Phoenix Lander Support

In addition to its science mission, MRO also serves as the latest member of an orbiting network of Mars relay spacecraft, providing telecommunications relay services to spacecraft on the Martian surface. This relay capability greatly increases the potential data return from small, energy-constrained landers and rovers, relative to what could be returned via direct-to-Earth (DTE) communications. For instance, to date the Mars Exploration Rovers have returned 98% of their data via UHF relay links, with only 2% returned via X-band DTE links^[8].

MRO provides relay services based on its Electra Proximity Link Payload^[9], consisting of dual-redundant Electra UHF transceivers, a quadrifilar helix UHF antenna, an internally redundant ultra-stable oscillator, a UHF coaxial switch and associated cabling. The Electra UHF transceiver represents a significant step beyond prior Mars relay payloads. It utilizes a software-defined radio architecture that allows extensive re-programmability of the radio functionality over the life of the MRO mission, enabling infusion of new algorithms and protocols, and allowing response to unforeseen mission scenarios. Electra also offers increased performance, supporting data rates from 1 kbps up to 1024 kbps (4 times higher than the UHF transceivers on Odyssey and MER) and frequency-agile multi-channel operation (unlike prior fixed-frequency Mars UHF transceivers.)

Although occurring prior to the formal start of the Relay phase, the 2007 Phoenix Lander mission represented the first operational relay customer for MRO. (While the Spirit and Opportunity rovers were still operational when MRO arrived, nominal MER relay support was provided by Odyssey, in order to minimize impact of MRO's primary science phase, aside from a handful of MER-MRO demonstration passes scheduled to verify MRO Electra performance.) Based on the success of relay communications in support of MER, the Phoenix mission chose to de-scope the lander's X-band direct-to-earth telecommunications capability in order to reduce landed mass and mission cost; thus, after separation from its cruise stage shortly before Mars atmospheric entry, all communications to and from the Phoenix Lander had to be conducted via the available relay orbiters: Odyssey, Mars Express, and MRO.

The first key relay function was to provide critical event communications support during Phoenix Entry, Descent, and Landing (EDL) on May 25, 2008. The purpose of this support was to capture sufficient engineering telemetry to support diagnosis of any potential mission failure during this critical mission phase. The high latitude Phoenix landing site enabled all three orbiters, each in near-polar orbit planes, to view the EDL trajectory. For this EDL communications function, MRO's Electra payload offered a unique capability to acquire a high-fidelity open-loop recording of the Phoenix signal: instead of attempting to demodulate the Phoenix signal in real-time, Electra instead acquired quadrature samples of the received Phoenix UHF signal, with a bit depth of 8 bits per sample, and a sampling rate of 150 kHz, throughout EDL. After landing, the open-loop recorded data were transmitted to Earth, where post-processing was performed to detect and track the Phoenix carrier signal and to demodulate the Phoenix telemetry bit stream. This strategy offered an extremely robust approach to EDL communications. In the event of an anomaly during EDL, a conventional real-time demodulation approach might suffer a loss of lock due to unexpected signal dynamics; by contrast, the open-loop strategy ensured a complete signal recording that could be processed at length after the fact to extract much greater information if an anomaly had occurred.

As an aside, the Phoenix EDL recording also allowed for a superb opportunity to provide additional performance data of the EDL. Using extremely careful planning, the spacecraft operated its HiRISE camera and was able to slew across the path of Phoenix as it hung on its parachute. Figure 7 is a testament to the flight team's ingenuity and professionalism, the first ever image of an EDL on another planet verified the performance of the parachute system.

Once on the surface, Odyssey and MRO provided multiple relay sessions per sol for delivery of commands to, and return of telemetry from, the Phoenix Lander. Because of ongoing science activities, MRO's level of relay support for Phoenix was limited to a single pass per day. During emergency situations, additional relay passes were added in order to meet the needs of the lander. Over the course of the Phoenix surface mission, which ended after 151 sols, MRO supported 252 relay passes, returning 6.8 Gb of Phoenix telemetry, and an additional 1.1 Gb of Phoenix fill data (corresponding to times where the relay capability exceeded the volume of Phoenix queued data). In combination with Odyssey, the two orbiters averaged over 250 Mb/sol of data return, more than 4 times the Phoenix requirement of 60 Mb/sol.



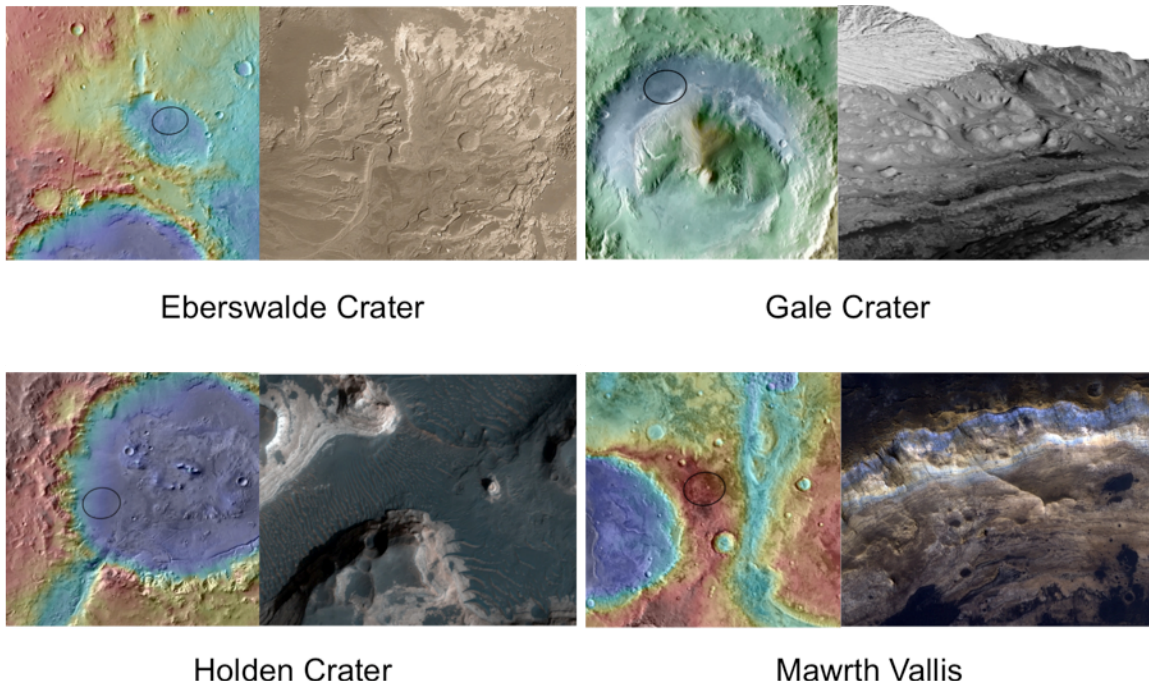
Figure 7. Phoenix descending on its parachute
(Credit University of Arizona, Lockheed Martin Space Systems, JPL, and NASA)

MRO has overcome some significant obstacles in the course of providing this Phoenix relay support. Several of the MRO science instruments had difficulty meeting the spacecraft radiated emission requirements in the UHF band in which Electra operates. The resulting electromagnetic interference (EMI) from those instruments had the potential to significantly degrade Electra's UHF receiver performance or even to prevent communications by causing Electra to lock onto the spurious signals rather than the intended Phoenix signal. This situation was known prior to launch, and the baseline solution was to simply power off the non-compliant science instruments. However, such power cycling would have represented a risk to the instruments and would have significantly impacted MRO science operations. To reduce these impacts, Electra's software-reconfigurable architecture was leveraged after MRO launch to upload modifications to the transceiver's signal processing algorithms, implementing a digital filter that attenuated the instrument EMI to an acceptable level.

In addition to the digital filter, MRO's instruments were put into operational modes (without power cycling) that would minimize interference with the relay link. The nominal MRO relay configuration has the spacecraft nadir-oriented, CRISM not observing, SHARAD not transmitting, and MCS paused at a particular attitude to minimize EMI.

Reconnoitering for MSL - Landing Site Scout

In addition to its programmatic telecommunications relay tasks, MRO performs the vital function of reconnoitering future landing sites. In this capacity, MRO's science instrument suite is used to acquire information about candidate locations. This data aids program planners in identifying, characterizing, and certifying candidate sites. Identification means the ability to find a potential new landing site of interest, typically using data acquired by CTX, CRISM, HiRISE, and possibly SHARAD.



- Eberswalde Crater (upper left) has the best deltaic formation (Mississippi or Nile deltas analog) yet found on Mars, clearly indicating several episodes of massive flow from a channel to the west.
- Gale Crater (upper right) has a mound in its center higher than the crater rims. The ground-level perspective view was created using a Digital Terrain Map developed from a HiRISE stereo pair.
- Holden Crater (lower left) has a massive channel breaching the southwest rim and the HiRISE image shows the stratified step that erosion has formed in the crater.
- Mawrth Vallis (lower right) has the target areas in the landing site ellipse; the mineral diversity and stratification are attractive features of this site.

Figure 8. Four Candidate MSL landing sites
(Credit University of Arizona, JHU Applied Physics Laboratory, JPL, and NASA)

Characterization refers to the application of data acquired using any or all instruments in the MRO suite to understand the scientific potential of the site for future discovery and to provide initial assessments of site safety. Certification refers to the final engineering and safety certification of the final landing site, typically requiring complete HiRISE stereo coverage, but also benefiting from CTX and CRISM data for traverse planning and MCS profiles for EDL design and simulation. These data are inputs to the certification process of the requesting mission, within the context of that mission's specific technical needs.

By December 2010, MRO had completed its coverage of the four prime candidate MSL landing sites. Figure 8 shows the finalists for the MSL landing in August 2012. The colored panels show the topography (red = high; purple = low) of the various sites. In addition, a candidate MSL landing ellipse is shown. Three of the four sites are "go to" sites, meaning that the rover drives out of the landing ellipse to get to the scientifically most interesting places.

4. FLIGHT AND MISSION OPERATIONS CHALLENGES

Overall, the MRO spacecraft has performed superbly over its prime mission and the Project is taking a fairly healthy spacecraft into its first Extended Mission. This mission has successfully demonstrated valuable engineering payloads (surface/orbit telecommunications with the Electra UHF transceiver; high data rate, low power Ka-Band telecommunications; and approach navigation with the Optical Navigation Camera) along with its suite of scientific payloads that have produced groundbreaking scientific discoveries. These successes have been achieved despite a handful of engineering and instrument challenges. Future successes are fully expected despite the need to continue to deal with these challenges -- challenges that constrain how we operate the spacecraft and challenges that will cause unplanned interruptions in MRO's operations.

In regards to the spacecraft bus, the flight team has been challenged by two issues within the telecommunications subsystem, by periodic star tracker resets and by a series of unexplained command and data handling (C&DH) subsystem resets.

There are two issues within the telecommunications subsystem: 1) the Ka-band exciter of the small deep space transponder (SDST-1) failed; and, 2) an apparent stuck waveguide transfer switch preventing swapping between traveling wave tube assemblies (TWTAs). After 8 months of successful operations (6000+ hours), the Ka-band exciter failed. After an extensive review, it is believed that the unit was overstressed during pre-launch testing. Since each of the two SDSTs on MRO would have been subject to the same stresses during ground testing, the second Ka-band unit (on SDST-2) is believed to be vulnerable to the same

failure. Because of what was achieved during the Ka-band demonstration and because of the significant system reconfiguration needed to go to SDST-2, the Project ended the Ka-band demonstrations. The Project still has the option to resume the demonstration or to deploy the Ka-band as our primary telecommunications in the event of problems with the X-band system or in the event of further problems with the waveguide transfer switch. The problem with the waveguide transfer switch is that the switch is believed to have become stuck in an intermediate position. Based upon the performance with the current switch position, the switch is assessed to be just slightly off alignment with the current configuration allowing us to use the TWTA-2 to conduct our nominal (HGA – high gain antenna) mission but at a slightly degraded power level (~1dB) and allowing us to use TWTA-1 for safe mode (or any desired LGA – low gain antenna) activities. After review, an investigation team recommended that no further attempts be made to move the switch as long as the telecommunications system remains healthy. The Project will continue to operate the spacecraft with the switch in its present configuration unless some very compelling circumstance indicates otherwise.

The issue with the star tracker is not viewed as a serious concern. The unit is known to be sensitive to single event upsets (SEUs) and periodically suffers a reset. The spacecraft operations team has developed a contingency plan for such an event and the recovery is very swift with no impact to the operation of the spacecraft and with minimal impact on the operations team in the recovery effort.

The final operational issue on the engineering side of the spacecraft is a series of unexplained resets of the C&DH system. In 2009, the spacecraft experienced four system resets. The resets appeared to be increasing in frequency and in severity. After the fourth reset, the Project held the spacecraft in safe mode for 111 days while a fix to a potentially serious vulnerability was developed that was revealed in the fourth reset. During that time, no further resets were experienced. In fact, the spacecraft operated without another such reset for over a year. In September 2010, a related reset occurred, but as of this writing no further resets have occurred. After a very extensive investigation, the leading candidate for the series of resets is still believed to be an SEU. The team has developed contingency plans for recovery of the spacecraft and is able to quickly return the spacecraft to service. Latest procedural updates allow for the recovery of the spacecraft in a 24 to 36 hour time period and a return to science operations in a 72 hour time period.

On the instrument side of the spacecraft, four of the six instruments have experienced challenges. The CRISM Team has had cooler and gimbal problems, the SHARAD Team has experienced occasional resets, the MCS Team has experienced gimbal issues and the HiRISE Team has experienced degradation as a result of chlorine contamination in the analog-to-digital converters (ADCs).

Each one of the respective science operations teams has developed operational workarounds to mitigate the particular problems that they have encountered. Updates to temperature set points, imposition of range of motion restrictions, etc., have allowed all of the instruments to continue to acquire meaningful scientific data. Additional discussion on the MRO challenges can be found in References [10] and [11].

Spacecraft and Instrument Status

Despite the challenges noted above, MRO is a very healthy spacecraft and enters its first Extended Mission in a good state. With the constraints of the telecommunication subsystem and the possible nuisance of future system resets, the engineering subsystems are fully operational and are performing strongly. The MRO engineering system was designed with long lifetimes and with significant redundancy. MRO has over 280 kg of usable propellant remaining and is able to support small orbit changes for future missions if required. Despite the challenges faced by the instruments, continued scientific gains from all instruments are fully expected throughout the first Extended Mission and beyond.

5. EXTENDED MISSION – OBJECTIVES & PLANS

The MRO Mission was recently approved for a two-year extension by NASA. The science and programmatic (mission support) objectives for MRO for this first Extended Mission are largely the same as for the Extended Science Phase and the original Relay Phase. For science, the EM objectives are defined by the acquisition of data from a third Mars year for interannual comparisons, by the specific targets and the expanded regional coverage of stratigraphic and compositional units that will test the hypotheses formed on the basis of the PSP and ESP observations. For relay, this reflects the slip of the Mars Science Laboratory (MSL) launch from 2009 to 2011, with MSL arrival in the summer of 2012.

The distinctive attributes of MRO include: 1) the ability to observe with higher spatial resolution than other orbiters in each of 3 observational categories—atmospheric monitoring, surface/subsurface survey, and coordinated observations of specific targets; and 2) an iterative planning process that optimizes observations by multiple instruments in their various observing modes. The spacecraft, flight software (specifically an onboard integrated targeting algorithm), mission design (e.g., orbit), ground data systems and project teams have—and continue to—exploit these capabilities. Major advances in our understanding of Mars have been derived from the unprecedented volumes of data returned to Earth.

Science Objectives

For science, MRO will continue to pursue its “Follow the Water” strategy. Its science goals for the Extended Mission (EM) are largely the same as for the ESP. To achieve the objectives for an extended mission, the MRO Project will continue to operate the spacecraft as it is currently done, with all instruments (except ONC) observing in well-defined and tested modes, and utilizing the well-established procedures of the present planning and commanding process. In a cost-constrained environment, operational changes are expensive because they require software development, testing and training to avoid being disruptive to the acquisition of new data. Table 2 shows the traceability matrix for the MRO science instruments to the EM science goals.

Programmatic Objectives

Unlike its prime mission, the MRO Extended Mission will emphasize support for MSL and landing site selection activities for future missions (including and beyond MSL). Relay operations use the MRO Electra package, while other mission support functions require use of the MRO science instruments. The contributions of these instruments and the missions benefiting from them are shown in Table 3.

Due to the dwindling fuel supply of ODY, MRO is slated to be the primary relay link for MSL and possibly 2016. Testing of the MRO Electra system at the data rates to be used by MSL is being conducted this year, but further testing once MSL is launched will likely occur. Relay will have priority during the MSL mission, and the MRO payload will be placed during relay passes in its nominal relay configuration: spacecraft nadir-oriented, CRISM not observing, SHARAD not transmitting, and MCS paused at a particular attitude—see Table 3. This mainly affects MRO science observations near the MSL site. However, as was done for PHX, it is expected that coordinated MRO and MSL observations, jointly agreed upon by the two projects, will be conducted.

During the EM MRO would cover landing sites for projected future missions such as the 2016 ESA EDL Demo Module (with its PHX-sized error ellipse) and the 2018 NASA-ESA EXM/MAX-C Mission (with its MSL-sized error ellipse). Since the 2018 NASA rover is projected to be a sample-caching rover, the 2018 site would have a high probability of being the site for the proposed Mars Sample Return Lander/Rover/Ascent Vehicle (MSR-L) that, in the current thinking, would return the sample cache from Mars to Earth through rendezvous in Mars orbit with an Earth Return Vehicle. Since the proposed sample return has both geochemical and astrobiological objectives, it is quite

Table 2. MRO Science Traceability Matrix

<i>Extended Mission Science Traceability Matrix</i>		HIRSE	CRISM	CTX	SHARAD	MCS	MARCI	Gravity
Science Goals	Science Objectives							
<i>Determine the nature and history of the Martian upper crust, emphasizing crustal stratigraphy and aqueous deposits</i>	Upper crust: determine structure, stratigraphy, mineralogy (esp. aqueous), and lithology (volcanic, sedimentary, metamorphic, or impactite).							
	Map deposits that show evidence for formation in surface water							
	Constrain where the crust has been influenced by groundwater							
	Determine what styles of volcanism, tectonism, and impact cratering have occurred and are there hydrothermal deposits?							
<i>Investigate the polar caps and layered terrains and ice at all latitudes</i>	Constrain the origin and evolution of the Polar Layered Deposits (PLD)							
	Determine the behavior of the seasonal and residual H ₂ O and CO ₂ polar ice deposits and how these affect present-day climate.							
	Examine mid-latitude water ice occurrence and behavior and how these provide crucial clues to longer-term climate variations.							
<i>Characterize ongoing surface changes, including aeolian processes, slope processes such as gullies, and impact cratering</i>	Determine the current impact rate, physical properties of the impacting objects, and implications for rates of ongoing or recent processes.							
	Estimate the current rate and processes of gravity-driven erosion of sloping terrain, investigating H ₂ O, CO ₂ and dry flow related processes.							
	Determine if dunes and ripples are forming and migrating in the present-day climate or if are they largely static.							
	Search for and examine ongoing processes in ice-rich ground.							
	Search for evidence of current activity such as tectonism or volcanism that could produce significant atmospheric methane.							
<i>Capture atmospheric interannual variability, while extending the climatological record</i>	Records of atmospheric weather, thermal structure, dust, ice and water vapor							
	Interannual variability							
	Atmospheric (vertical) structure: dust, water ice cloud, particle properties							
	Atmospheric dynamics: tides and waves							

Primary Contribution

Secondary Contribution

Table 3. MRO Programmatic Traceability Matrix

MISSION SUPPORT FUNCTIONS	MRO Programmatic Traceability Matrix	HIRSE	CRISM	CTX	SHARAD	MCS	MARCI	Gravity	MER	MSL	MAVEN	2016*	2018/MSR*
	OBSERVATION AND ANALYSIS PRODUCTS REQUIRED												
Landing Site Activities	Site identification, characterization and certification	REQUIRED INSTRUMENTS							MISSIONS SUPPORTED				
	Identification of sites with high potential for scientific return								✓			✓	
	Characterization of sites with respect to specific scientific and technical risk criteria								✓		✓	✓	
	Certification of site safety for specific mission assets (rovers/landers)								✓		✓	✓	
Environmental Monitoring	Environmental Data for Design and Planning, Near-real Time Coverage of Critical Events, Surface Operations	REQUIRED INSTRUMENTS							MISSIONS SUPPORTED				
	Environmental data for mission design and planning (e.g., atmospheric state; surface properties; gravity field)								✓	✓	✓	✓	
	Near-real time coverage of critical events (e.g., EDL and approach): atmospheric state (including dust activity); mission deployments/EDL								✓ E D L	✓ A B ^{**}	✓ A B /E D L	✓ E D L	
	Environmental monitoring for surface operations: Dust storms affecting power generation and visibility								✓			✓	
	Traverse planning								✓	✓		✓	
	EDL = entry/descent/landing; AB = aerobraking	Green => Primary Contribution Tan => Secondary Contribution							**MAVEN "deep dips"				
Relay (Electra)	Nominal state during UHF relay: OP = Operating; NT = Not Observing (CRISM)/Transmitting (SHARAD); FA = frozen at particular scan position	OP	NT	OP	NT	FA	OP		✓	✓		✓	✓

*proposed mission

possible that these later missions would go to a new site, possibly one in addition to those considered previously for MSL. Since no other mission currently flying, in development, or now being planned (including the proposed 2016 ESA-NASA orbiter mission) has the capabilities of MRO, especially with regard to the HiRISE spatial resolution needed for site certification, other programs are already starting to place high priority on MRO acquiring the desired data.

Extended Mission Timelines

The timeline for the MRO Extended Mission is shown in Figure 9. Although the EM boundaries are formally aligned

with the government's fiscal year (FY) for FY11-FY12, it is convenient for planning purposes to show the somewhat longer period between solar conjunctions. In this timeline the MRO propulsive maneuvers (burns) required to move to the earlier local times needed to cover MSL EDL are shown, along with curves showing the effect on both local mean and true solar times for MRO if the maximum change is required. The programmatic requirement for MRO is to cover MSL during EDL and its first use of the "skycrane" in 2012 during late northern summer (on both Earth and Mars) when MSL arrives at its destination. The timeline also shows seasonal periods, calendar dates, the projected data volume acquisition, and other mission milestones.

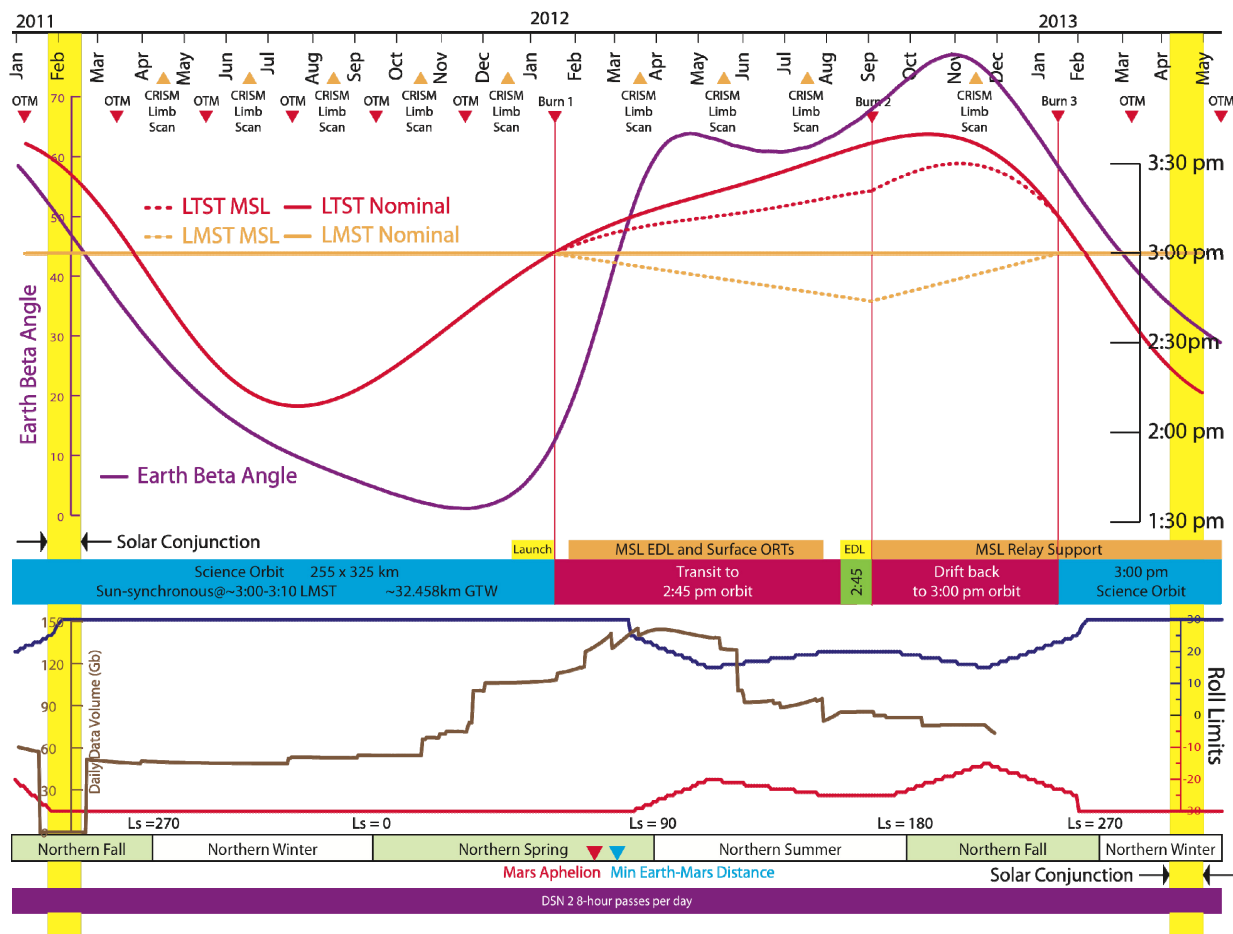


Figure 9. MRO Extended Mission Timeline

Relay Preparations for MSL

As noted earlier, MRO will be the prime relay asset for the MSL rover Curiosity. In this capacity, it is envisioned that MRO will be supporting two and up to three relay overflights per day. In support of the rover's surface operations, several enhancements are being made to MRO's relay capabilities. The first of these is a software update to the Electra programmable radio. This software update, known as Adaptive Data Rates (ADR), will allow MRO's Electra telecommunications package to communicate with landed vehicles at variable data rates during a single overflight pass. The second update is the use of existing onboard (flight software) capability that will allow the MRO spacecraft to roll off-nadir (target) during relay passes.

Electra Adaptive Data Rates (ADR) - The ADR capability allows the MRO spacecraft and the MSL rover to autonomously alter the data rate they are using for their relay communication session as the overflight pass progresses. In comparison, MRO's support of the MER rovers and the Phoenix lander used a fixed data rate for the duration of an overflight. The choice of the "right" data rate for these missions was made based on the quality of the overflight geometry and the desired data volume transfer. These teams could choose a low data rate which would mean that they would establish the communication link earlier and hold it for a longer duration but they would only receive a small volume of data due to the low data rate. Alternatively, these teams could choose a high data rate which would mean that they would establish the communication link later and hold it for a shorter duration but again, they would only receive a relatively small data volume. These missions would work to identify a best data rate for a given pass in order to obtain the best data return for a fixed data rate. In contrast, a greater volume of data would be obtained by starting a pass as soon as possible with a lower data rate and then progressively stepping up to higher data rates as the geometry of the pass allowed and then progressively stepping down the data rate as the quality of the geometry degraded. This capability, being developed by the MSL Project, is expected to greatly enhance the data return of the rover Curiosity. A flight software update for the Electra radio will be required to implement this feature.

Spacecraft Rolls During Relay Overflights (Rolls with Relay) - The second enhancement to MRO's relay capabilities is the ability to perform rolls while engaged in a relay overflight. Recall that the current baseline MRO configuration for a relay pass has the spacecraft nadir oriented. The ability to roll (ITL driven) during a relay pass is not a limitation in the MRO flight software; it is rather a limitation in ground planning capability and tools. Motivated by the desire to enhance the relay communications link and to image off-nadir during relay passes, the MRO Project has redesigned elements of its flight operations to permit rolls during relays. This enhancement allows MRO to roll so that its UHF antenna can be pointed towards the rover and thus increase data rates

for the overflight (relay-driven rolls). Alternatively, when relay overflight geometry is within prescribed tolerances or the relay link is not being utilized, the spacecraft can be rolled allowing its imagers to observe targets in the vicinity of the landed vehicle (science-driven rolls).

Integration of Relay into the Science Operations Process (Dual Mission) - One of the fundamental assumptions in developing the ability to support relay-driven rolls and to support science-driven rolls during relays was that the requests for rolls and the requests for relays would integrate with MRO's existing operational processes and timelines.

Figure 10 illustrates the general process flow for integrating relay requests, payload observation requests, and engineering activities with the ability to perform rolls with relays. This process flow is essentially identical to the process flow as it was before the introduction of the recently developed rolls with relay capability. Prior to the rolls with relays capability, all relay passes (with the MER rovers and with the Phoenix lander) were nadir-pointed relays. With this new capability, there is a new path in the process performed by the Electra operations team that defines the allowable UHF roll range ("Allowed Roll Range"). There is also a loop for Mission Planning and Sequencing Team (MPST) to check compliance of the roll range and to give their 'go' that was absent in the previous process.

As noted in the diagram, the figure only shows the core process and does not provide the iterations for the Interactive Observation (IO) and the Non-Interactive Observation (NIO) versions of the Payload Target File (PTF) used to define the payload observations for a given two-week operational sequence. Briefly, the IO PTF covers the full duration of the two-week operational sequence and contains heavily coordinated, off-nadir and/or high-stability payload observations, and the NIO PTFs cover one-week segments of the larger operational sequence and contain payload observations that do not interfere with the operation of other payloads. These iterations are part of the larger operational practices and are not covered in this paper.

To initiate the process illustrated in Figure 10, the navigation team will deliver information detailing the overflights available that will allow orbiter/rover relay communication sessions. This information is delivered to the Electra Operations Team (EOT) and to the MSL Operations Team via a JPL environment called MaROS (Mars Relay Operations Service). The EOT uses the navigation information to assess the quality of the available relay passes and to determine the allowable roll angle range for each of those passes. The allowable roll angle range provides the range of roll angles that would support the successful transfer of the desired quantity of data over that overflight. The MSL Operations Team uses the navigation information to determine what data volume they might expect for a given overflight and to determine if they want to request a roll to increase the available data volume.

Next, MSL provides their relay requests to MRO's MPST using an Overflight Request File (ORF -- previously known as an APGEN Plan File) via the MaROS system and EOT provides MPST with the Allowed Roll Range. MPST takes this information and generates the Relay PTF. MPST provides the Relay PTF to the Payload Operations Support Team (POST) who merges this PTF with the PTFs from the payload teams to create the Integrated Payload Target File (IPTF). POST then sends the IPTF to the MPST for them to perform a roll range compliance check and thus confirm that all requested rolls during relays conform to the allowed roll range. If violations are found, the teams would need to perform another iteration to eliminate the violation. Upon successful completion of this confirmation activity, MPST

gives its 'go' to POST to release the IPTF to the Flight Engineering Team (FET). The FET takes this IPTF and generates the Integrated Target List (ITL). The ITL is the list of observations and relays to be performed by the spacecraft over a given operational sequence.

The MRO team has successfully evolved their existing, very compressed science and relay planning process and schedule to adapt to the expected heavy demands of the MSL project while increasing their ability to continue to perform a high-value science mission.

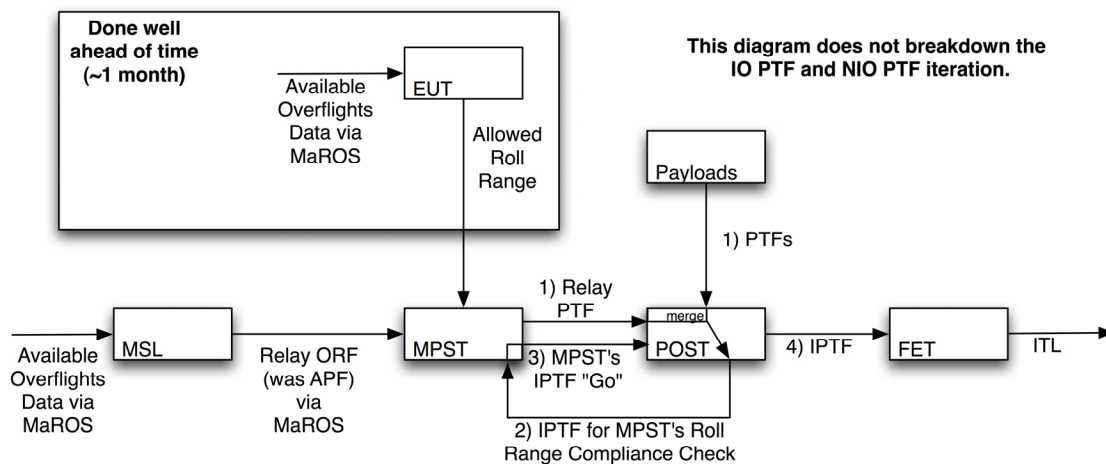


Figure 10. Process Flow for Target File Generation

6. SUMMARY

MRO is a flagship mission of the Mars Exploration Program. Its powerful suite of science instruments continues to unveil Mars in unprecedented detail. The MRO science investigations show that Mars is a diverse planet with a complex geologic history. In particular, the diversity of early water-rich environments shows preservation potential for signatures of ancient life, if it developed. Notable science achievements from the prime mission (September 2010) include:

- Over 123 Terabits of Science Data
- Acquisition of over 90,000 images, and image equivalents
 - 62% of the planet imaged at 6 m/pixel (CTX)
 - 1.2% imaged at 30 cm/pixel (HiRISE)
 - 70% of planet covered in 72 spectral channels at low opacity (CRISM)
 - Daily Global Maps (MARCI)
 - Atmospheric Profiling (MCS)
 - Shallow, Subsurface Soundings (SHARAD)

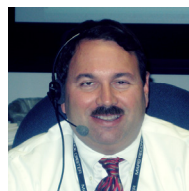
In addition to its science achievements, MRO significantly enhanced the safety of the landing phase of the Phoenix mission by both identifying that the previous prime landing site was too rocky for safety, and by identifying safe locations for use, ultimately including the final chosen (and successful) site.

As MRO prepares for its Extended Mission, spacecraft capabilities and operations processes are being refined for its dual mode mission of science platform and relay satellite. During the EM, MRO will continue to return unparalleled volumes of data – either from its own suite of instruments or from other surface missions. Preparations are well underway for the anticipated new demands that the MSL Rover Curiosity will place on the MRO spacecraft and its operations teams when it arrives in August 2012. With significant redundancy and large fuel reserves still remaining, the MRO spacecraft is expected to stay on-orbit for many years to come.

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BIOGRAPHY



Dan Johnston received a B.S. in Aerospace Engineering from the University of Texas in 1984 and an MSE from the University of Texas in 1989. Since joining JPL in 1989, he has participated in the development and flight operations phases of the Mars Observer and Mars Global Surveyor missions. Prior to joining JPL, he was employed with McDonnell Douglas Astronautics in Houston, TX, in support of STS (Shuttle) rendezvous flight planning. Currently, Mr. Johnston is the Deputy Project Manager (and as additional duty the Mission Manager) of the Mars Reconnaissance Orbiter Project.

Dr. Richard Zurek graduated from Michigan State University with a BS in Mathematics in 1969 and received his Ph.D. in Atmospheric Sciences from the University of Washington (Seattle) in 1974. Following one-year post-doctoral appointments at the National Center for Atmospheric Research and at the University of Colorado in Boulder, Colorado, he went to work at JPL, where he has been employed since 1976. Dr. Zurek is currently the Program Scientist for the Mars Exploration Program (MEP) and the Project Scientist for the Mars Reconnaissance Orbiter Project.



David E. Herman received his B.S. in Aerospace Engineering from San Diego State University and his M.S. in Aerospace Engineering from the University of Michigan-Ann Arbor. After some time with The Aerospace Corporation, he joined the Jet Propulsion Laboratory in 1996. He has worked operations on Cassini and MRO, and development on MER, X2000 and OSTM's WSOA. He spent nearly four years as the group supervisor of the Flight Engineering Group before taking his current position as Chief Engineer on MRO.

Charles (Chad)



Edwards, Jr. is the Chief Telecommunications Engineer for NASA's Mars Exploration Program at the Jet Propulsion Laboratory. Prior to this current assignment he managed the research and development program of NASA's Deep Space Network. He received an A.B. in Physics from Princeton University and a Ph. D. in Physics from the California Institute of Technology.

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